SW

SHAOFU WANG
MECHANICAL
ENGINEERING

PORTFOLIO

About Me

This portfolio highlights my representative projects across mechanical design, fluid dynamics and software development.

FIELDS

- Mechanical Design (CAD/FEA)
- Robotics & Control
- iOS Development
- Fluid Dynamics

PROJECT INDEX

01 Smart Cabin Lock — MECHANICAL
02 Oops Budget — SOFTWARE
03 Auto-Clean Window — ROBOTICS
04 Drag Reduction — FLUID ANALYSIS
05 Lab Assistant — ROBOTICS
06 CardCraft — SOFTWARE

PROJECT 01

Smart Bin Lock

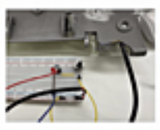
CONTEXT

A functional prototype for an automated luggage compartment lock, bridging mechanical hardware with electronic control.

ROLE
MECH DESIGN

SOFTWARE
CATIA

STATUS
PROTOTYPING



PLATFORM
iOS APP

STACK
SWIFT/SWIFTUI

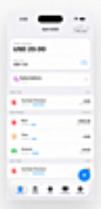
AI
GEMINI API

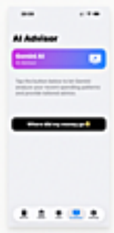
ABSTRACT

An iOS accounting app powered by Gemini AI, delivering personalized financial insights.

PROJECT 02

Oops Budget





PROJECT 03

Auto-Clean Window

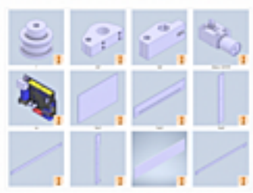
CONTEXT

An automated cleaning solution integrating mechanical, electronic, and software systems, designed to clean high-rise windows autonomously via remote control.

CATEGORY
ROBOTICS / IOT

CONTROL
ARDUINO UNO

ACTUATOR
JOYSTIK MOTOR



SUBJECT
FLUID DYNAMICS

METHOD
CFD/ CAM

ANALYSIS
1ST PRESS SIM

ABSTRACT

Investigated ventilated cavity technology to significantly reduce the underwater drag of a ship. Established formula via least-squares curve fitting.

PROJECT 04

Drag Reduction

PROJECT 05

Lab Assistant

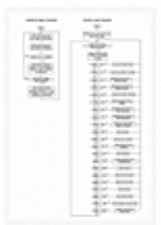
CONTEXT

A robotic assistant designed to conduct hazardous or inaccessible chemistry experiments remotely. Features a 6 DOF robotic arm and ESP32-CAM video streaming.

FIELD
ROBOTICS / IOT

BOARD
ARDUINO

NETWORK
ESP32



PLATFORM
iOS APP

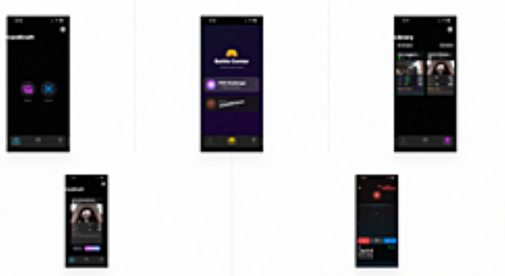
STACK
SWIFT/SWIFTUI

ABSTRACT

Turn everyday objects into Pokémon-style collectible cards with AI.

PROJECT 06

CardCraft



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THANK YOU

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